

Algebraic Roots & Indices

Four laws, three special powers, one way to solve.
Every indices question is built from these.



DR ESLAM AHMED
eliteigcse.com

THE TOPIC ON ONE MAP

Algebraic Roots & Indices

01 — The four laws

$$x^3 \times x^5 \quad (3x^2)^3 \quad \frac{12a^4h^6}{4ah^2}$$

02 — Zero and negative powers

$$g^0 = 1 \quad x^{-2} = \frac{1}{x^2}$$

03 — Fractional powers

$$8^{\frac{2}{3}} \quad \text{Flip} \rightarrow \text{Root} \rightarrow \text{Power} \quad \text{"exact value"}$$

04 — Solving: unify the base

$$2^n = 8^3 \quad \text{"find } n\text{"} \quad \text{"x in terms of } n\text{"}$$

05 — Where you will use it

$$\text{"in the form } ax^n\text{"} \quad \text{differentiation}$$

01 The Four Laws

Same base only. Numbers in front are worked out normally — the laws are for the powers.

WHEN YOU SEE → YOU DO

“simplify” with the same base

→ $\times$ add $\cdot \div$ subtract $\cdot ()^n$ multiply

“(3x²)³”

→ the number gets the power too

Multiply

$$a^m \times a^n = a^{m+n}$$

add the powers

Divide

$$a^m \div a^n = a^{m-n}$$

subtract the powers

Power of a power

$$(a^m)^n = a^{mn}$$

multiply the powers

Power of a product

$$(ab)^n = a^n b^n$$

everything inside gets the power

WORKED MODEL • NUMBERS AND LETTERS

$$\frac{36af^8}{12a^5f^2}$$

1 $36 \div 12 = 3$

numbers: normal

2 $a^{1-5} = a^{-4}$ $f^{8-2} = f^6$

letters: subtract

ANSWER $\frac{3f^6}{a^4}$

× **TRAP** $(3x^2)^3 = 3x^6$ is wrong — the 3 is inside the bracket too: $3^3x^6 = 27x^6$.

IDEA 01

Practise: The Four Laws

My questions first, then selected practice, then real past papers.

Q1 DR ESLAM One law each

CH 2 · TOPIC 2 3 marks

Simplify (a) $x^5 \times x^3$ (b) $y^9 \div y^4$ (c) $(x^4)^3$

Q2 DR ESLAM Numbers inside the bracket

CH 2 · TOPIC 2 2 marks

Simplify $(2x^3y)^4$.

The 2 gets the power too: $2^4 = 16$.

Answer _____

Q3 **DR ESLAM** Numbers, then letters

CH 2 · TOPIC 2 2 marks

Simplify $\frac{12x^7y^2}{4x^3y^5}$.

Answer _____

SELECTED PRACTICE**Q4** **PRACTICE** Practice · Easy

CH 2 · TOPIC 2 3 marks

2 (a) Simplify $m^5 \div m^3$

(1 mark)

(b) Simplify $5x^4y^3 \times x^2y$

(2 marks)

Q5**PRACTICE**

Practice · Easy

CH 2 · TOPIC 2 1 mark

13 Simplify.

$$a^6 \div a^2$$

(1 mark)

Q6**PRACTICE**

Practice · Medium

CH 2 · TOPIC 2 2 marks9 Simplify $2a^3b \times 5a^2b^3$

(2 marks)

Q7**PRACTICE**

Practice · Hard

CH 2 · TOPIC 2

2 marks

13 $(ar^b)^4 = 16r^{20}$ where a and b are positive integers.

Work out a and b

$a = \dots\dots\dots$

$b = \dots\dots\dots$

(2 marks)

PAST PAPERS

Q8**PAST PAPER**

Edexcel 4MA1 · January 2020 · Paper 2H · Q1

CH 2 · TOPIC 2

3 marks

You must write down all the stages in your working.

1 (a) Simplify $\frac{x^9}{x^2}$

(1)

(b) Write $\frac{7^8 \times 7^4}{7^3}$ as a single power of 7

(Total for Question 1 is 3 marks)

Q9**PAST PAPER**

Edexcel 4MA1 · November 2021 · Paper 2H · Q1

CH 2 · TOPIC 2 4 marks

- 1 (a) Write down the value of m , given that $3^4 \times 3^5 = 3^m$

 $m = \dots\dots\dots$

- (b) Write down the value of n , given that $(5^3)^7 = 5^n$

 $n = \dots\dots\dots$ **(1)**

- (c) Find the value of p , given that $\frac{7^8 \times 7^2}{7^p} = 7^6$

 $p = \dots\dots\dots$ **(Total for Question 1 is 4 marks)**

Q10**PAST PAPER**

Edexcel 4MA1 · January 2022 · Paper 1H · Q1

CH 2 · TOPIC 2 6 marks**You must write down all the stages in your working.**

1 (a) Simplify $a^7 \times a^4$

.....

(b) Simplify $w^{15} \div w^3$

.....

(c) Simplify $(8x^5y^3)^2$

.....

(d) Make t the subject of $c = t^3 - 8v$

.....

(Total for Question 1 is 6 marks)

Q11**PAST PAPER**

Edexcel 4MA1 · May 2023 · Paper 1HR · Q6

CH 2 · TOPIC 2 4 marks6 (a) Simplify $m^{10} \div m^3$

.....

$$k^n \times k^4 = k^{12}$$

(b) Write down the value of n $n =$ (c) Simplify $(3x^6y^8)^2$

.....

(Total for Question 6 is 4 marks)

Q12**PAST PAPER**

Edexcel 4MA1 · November 2023 · Paper 2H · Q4

CH 2 · TOPIC 2 8 marks

4 (a) Solve $\frac{2x+5}{6} = 2x-5$

Show clear algebraic working.

$$x = \dots\dots\dots$$

(b) Simplify $h^{15} \div h^3$

$$\dots\dots\dots$$

(c) Simplify fully $(2g^3k^5)^4$

$$\dots\dots\dots$$

(2)

(d) Given that $\frac{y^5 \times y^n}{y^7} = y^{12}$

work out the value of n

$$n = \dots\dots\dots$$

(Total for Question 4 is 8 marks)

Q13**PAST PAPER**

Edexcel 4MA1 · June 2025 · Paper 1H · Q4

CH 2 · TOPIC 2 4 marks

4 $x^5 \times x^7 = x^m$

(a) Find the value of m

$m = \dots\dots\dots$

$y^8 \div y^3 = y^n$

(b) Find the value of n

$n = \dots\dots\dots$

(c) Simplify fully $(5a^4r^2)^3$

$\dots\dots\dots$

(Total for Question 4 is 4 marks)

02 Zero and Negative Powers

Zero means 1. Negative means **flip it**.

WHEN YOU SEE → YOU DO

“write down the value of g^0 ”

→ 1

a negative power

→ flip it

0

Power zero

Anything (except 0) to the power 0 is 1: $g^0 = 1$, $(3x^2y)^0 = 1$, but $5y^0 = 5 \times 1 = 5$.

-

Negative power = flip

$$a^{-n} = \frac{1}{a^n} \quad 2^{-3} = \frac{1}{8} \quad \left(\frac{x}{y}\right)^{-2} = \left(\frac{y}{x}\right)^2 = \frac{y^2}{x^2}$$

A negative power never makes the answer negative — it moves the term across the fraction line.

WORKED MODEL • FLIP, THEN THE LAWS

$$\frac{3^{-2} \times 3^5}{3^{10}} = 3^n$$

$$1 \quad 3^{-2+5} = 3^3$$

top: add

$$2 \quad 3^{3-10} = 3^{-7}$$

divide: subtract

ANSWER $n = -7$

× **TRAP** $5y^0 \neq 1$. Only y has the power 0: $5y^0 = 5$. But $(5y)^0 = 1$.

IDEA 02

Practise: Zero and Negative Powers

My questions first, then selected practice, then real past papers.

Q14 DR ESLAM Zero and negative

CH 2 · TOPIC 2 3 marks

Work out (a) 5^0 (b) 2^{-3} (c) $\left(\frac{2}{3}\right)^{-2}$

Q15 DR ESLAM Negative = flip

CH 2 · TOPIC 2 1 mark

Write $\frac{1}{x^4}$ as a single power of x .

Answer _____

SELECTED PRACTICE**Q16** PRACTICE Practice · Easy

CH 2 · TOPIC 2 1 mark

1 Simplify $(m^{-2})^5$

(1 mark)

Q17**PRACTICE**

Practice · Easy

CH 2 · TOPIC 2 1 mark

9 Simplify $(3x^2y)^0$

(1 mark)

Q18**PRACTICE**

Practice · Easy

CH 2 · TOPIC 2 1 mark

11 Simplify $w^1 \times w^0$

Circle your answer.

1	0	w	w^2
---	---	-----	-------

(1 mark)

Q19**PRACTICE**

Practice · Medium

CH 2 · TOPIC 2 7 marks

8 (a) Solve $3x^2 = 147$

(2 marks)

(b) Work out the value of 2^{-3}

(1 mark)

(c) Simplify $(3x^2)^3$

(2 marks)

(d) $w = 4p - 16$

Make p the subject of this formula.

(2 marks)

PAST PAPERS

Q20**PAST PAPER**

Edexcel 4MA1 · January 2021 · Paper 2HR · Q4

CH 2 · TOPIC 2 4 marks

4 (a) Simplify $(3k^2)^4$

.....

(b) Simplify $(21m^4n) \div (3n^{-5})$

.....

(Total for Question 4 is 4 marks)**Q21****PAST PAPER**

Edexcel 4MA1 · June 2022 · Paper 1H · Q9

CH 2 · TOPIC 2 4 marks

9 (a) Simplify $8 \times (4t)^0$

.....

$x^6 \div x^{-5} = x^p$

(b) Find the value of p

$p =$

(c) Simplify fully $(2k^2m^4)^3$

.....

(Total for Question 9 is 4 marks)

Q22**PAST PAPER**

Edexcel 4MA1 · June 2023 · Paper 2HR · Q7

CH 2 · TOPIC 2 2 marks7 (a) Simplify $(4^{-2})^0$

$$3^{-14} \times 3^8 = 3^m$$

(b) Find the value of m

$$m = \dots\dots\dots$$

(Total for Question 7 is 2 marks)**Q23****PAST PAPER**

Edexcel 4MA1 · May 2024 · Paper 1H · Q10

CH 2 · TOPIC 2 4 marks10 (a) Simplify $(2p)^0$ where $p > 0$

$$y^9 \times y^{-3} = y^n$$

(b) Find the value of n

$$n = \dots\dots\dots$$

(c) Simplify fully $(5a^4c^2)^3$ **(Total for Question 10 is 4 marks)**

03 Fractional Powers

The **bottom** of the fraction is the root. The **top** is the power.

WHEN YOU SEE → YOU DO

a fractional power

→ bottom = root, top = power

“simplify fully $(\dots)^{-\frac{2}{3}}$ ”

→ Flip → Root → Power

“exact value”

→ no calculator decimals — keep fractions

$$a^{\frac{m}{n}} = \left(\sqrt[n]{a} \right)^m$$

root first — the numbers stay small

F

Flip

If the power is negative, flip the fraction and make the power positive.

R

Root

Take the root of **every** part: numbers, and divide each letter's power.

P

Power

Raise to the top number.

WORKED MODEL • FLIP → ROOT → POWER

$$\left(\frac{64x^6}{25y^2}\right)^{-\frac{1}{2}}$$

$$1 \quad \left(\frac{25y^2}{64x^6}\right)^{\frac{1}{2}}$$

flip

$$2 \quad \sqrt{25} = 5 \quad \sqrt{64} = 8 \quad y^{2 \div 2} = y \quad x^{6 \div 2} = x^3$$

root

$$\text{ANSWER} \quad \frac{5y}{8x^3}$$

IDEA 03

Practise: Fractional Powers

My questions first, then selected practice, then real past papers.

Q24 DR ESLAM Bottom is the root

CH 2 · TOPIC 2 3 marks

Work out (a) $8^{\frac{2}{3}}$ (b) $27^{-\frac{1}{3}}$ (c) $\left(\frac{4}{9}\right)^{-\frac{3}{2}}$

Q25 DR ESLAM Flip → Root → Power

CH 2 · TOPIC 2 3 marks

Simplify fully $\left(\frac{x^6}{64y^3}\right)^{-\frac{2}{3}}$.

Flip first. Then cube-root every part. Then square.

Answer _____

SELECTED PRACTICE

Q26

PRACTICE

Practice · Medium

CH 2 · TOPIC 2 2 marks

2 Simplify $(9x^8y^3)^{\frac{1}{2}}$

(2 marks)

Q27

PRACTICE

Practice · Hard

CH 2 · TOPIC 2 3 marks

3 (a) Simplify fully $(3e)^0$

(1 mark)

(b) Simplify fully $\left(\frac{64x^6}{25y^2}\right)^{-\frac{1}{2}}$

(2 marks)

Q28**PRACTICE**

Practice · Hard

CH 2 · TOPIC 2 3 marks

9 Simplify completely $\left(\frac{16w^8}{y^{20}}\right)^{-\frac{3}{4}}$

(3 marks)

Q29**PRACTICE**

Practice · Hard

CH 2 · TOPIC 2 4 marks

16 (a) Simplify fully $(32a^{15})^{\frac{3}{5}}$

(2 marks)

(b) Express $\left(\frac{1}{10x}\right)^{-3}$ in the form px^n where p and n are integers.

(2 marks)

PAST PAPERS

Q30**PAST PAPER**

Edexcel 4MA1 • January 2020 • Paper 1H • Q17

CH 2 • TOPIC 2 6 marks

17 (a) Show that $(6 + 2\sqrt{12})^2 = 12(7 + 4\sqrt{3})$

Show each stage of your working.

(b) Simplify fully $\left(\frac{27a^{12}}{t^{15}}\right)^{-\frac{2}{3}}$

(Total for Question 17 is 6 marks)

Q31**PAST PAPER**

Edexcel 4MA1 · January 2021 · Paper 1HR · Q12

CH 2 · TOPIC 2 9 marks

12 (a) Simplify $(16e^{10}f^6)^{\frac{1}{2}}$

.....
(2)

(b) Write $\frac{2x+1}{4} + \frac{x-2}{3}$ as a single fraction in its simplest form.

Given that $4^{k+3} = 16 \times 2^k$

(c) find the value of k .
Show your working clearly.

$k =$

(Total for Question 12 is 9 marks)

Q32**PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 2H • Q11

CH 2 • TOPIC 2 3 marks

11 Simplify fully $\left(\frac{9t^4w^9}{18t^6w^{10}}\right)^{-2}$

.....

(Total for Question 11 is 3 marks)

Q33**PAST PAPER**

Edexcel 4MA1 • May 2021 • Paper 2H • Q14

CH 2 • TOPIC 2 3 marks

14 Simplify fully $\left(\frac{9x^4}{16y^{10}}\right)^{-\frac{1}{2}}$

.....

(Total for Question 14 is 3 marks)

Q34**PAST PAPER**

Edexcel 4MA1 • June 2023 • Paper 2HR • Q11

CH 2 • TOPIC 2 2 marks

- 11 Express $\left(\frac{m^6 k^{10}}{25}\right)^{\frac{3}{2}}$ in the form $\frac{m^a k^b}{c}$ where a , b and c are integers to be found.

.....

(Total for Question 11 is 2 marks)

Q35**PAST PAPER**

Edexcel 4MA1 · June 2023 · Paper 2H · Q15

CH 2 · TOPIC 2 7 marks15 (a) Simplify fully $(32a^{15})^{\frac{3}{5}}$

(2)

(b) Express $\left(\frac{1}{10x}\right)^{-3}$ in the form px^n where p and n are integers.

(2)

(c) Solve $\frac{1-2y}{3} = \frac{4}{5} - \frac{2y-1}{2}$

Show clear algebraic working.

 $y =$ **(Total for Question 15 is 7 marks)**

Q36**PAST PAPER**

Edexcel 4MA1 · November 2023 · Paper 2H · Q12

CH 2 · TOPIC 2 4 marks

12 $2^7 \times 4^5 = 4^x$

(a) Calculate the value of x

$$x = \dots\dots\dots$$

(2)

(b) Simplify fully $(125p^6y^{24})^{\frac{2}{3}}$

.....

(Total for Question 12 is 4 marks)

Q37**PAST PAPER**

Edexcel 4MA1 · May 2024 · Paper 1HR · Q17

CH 2 · TOPIC 2 4 marks

17 (a) $\left(\sqrt[4]{k^{12}}\right)^5 = k^n$

Find the value of n

$n = \dots\dots\dots$

(1)

(b) Express $\frac{7}{2 - \sqrt{3}}$ in the form $\sqrt{c} + d$ where c and d are integers.

Show your working clearly.

 $\dots\dots\dots$

(Total for Question 17 is 4 marks)

Q38**PAST PAPER**

Edexcel 4MA1 · June 2025 · Paper 2H · Q15

CH 2 · TOPIC 2 3 marks

15 Simplify fully $\left(\frac{2a^3}{10a^7c^2}\right)^{-3}$

 $\dots\dots\dots$

(Total for Question 15 is 3 marks)

Q39**PAST PAPER**

Edexcel 4MA1 · November 2025 · Paper 2H · Q13

CH 2 · TOPIC 2 4 marks

13 (a) Simplify fully $\left(\frac{125x^6}{y^{15}}\right)^{\frac{2}{3}}$

.....
(2)

Given that $4^n = \frac{4^m}{64^p}$

(b) Express n in terms of m and p

$n = \dots\dots\dots$

(Total for Question 13 is 4 marks)

04 Solving: Unify the Base

Make both sides a single power of the **same base**. Then the powers are equal.

WHEN YOU SEE → YOU DO

“find the value of n / k ”, $2^{\dots} = \dots$

→ unify the base, equate the powers

“express x in terms of n ”

→ same method, answer contains n

1

Unify the base

Write every number as a power of one prime: $4 = 2^2$, $8 = 2^3$, $16 = 2^4$, $\sqrt{2} = 2^{\frac{1}{2}}$.

2

One power each side

Use the four laws until each side is $2^{\text{something}}$.

3

Equate the powers

If $a^x = a^y$ then $x = y$. Solve — usually a linear equation.

WORKED MODEL • JAN 2021 P1HR

$$4^{k+3} = 16 \times 2^k$$

$$1 \quad (2^2)^{k+3} = 2^4 \times 2^k$$

unify: base 2

$$2 \quad 2^{2k+6} = 2^{k+4}$$

one power each side

$$3 \quad 2k + 6 = k + 4$$

equate powers

$$\text{ANSWER} \quad k = -2$$

CALCULATOR CHECK Check with the calculator: $4^{(-2)+3} = 4$ and $16 \times 2^{-2} = 4$. Same → correct.

IDEA 04

Practise: Solving: Unify the Base

My questions first, then selected practice, then real past papers.

Q40 DR ESLAM Unify the base

CH 2 · TOPIC 2 2 marks

Solve $3^{x+1} = 81$.

$81 = 3^?$

$x =$ _____

Q41 DR ESLAM Both sides need changing

CH 2 · TOPIC 2 3 marks

Solve (a) $2^{3x} = 4^{x+1}$ (b) $9^x \times 3 = 27^2$

SELECTED PRACTICE

Q42 **PRACTICE** Practice · Hard

CH 2 · TOPIC 2 3 marks

$$1 \quad 16^{\frac{1}{5}} \times 2^x = 8^{\frac{3}{4}}$$

Work out the exact value of x .

(3 marks)

Q43 **PRACTICE** Practice · Hard

CH 2 · TOPIC 2 1 mark

$$5 \quad \text{Given that } \left(\sqrt[3]{\frac{1}{x}} \right)^4 = x^m$$

find the value of m $m = \dots\dots\dots$

(1 mark)

Q44**PRACTICE**

Practice · Hard

CH 2 · TOPIC 2 4 marks

8 Given that $4^{k+3} = 16 \times 2^k$

find the value of k .

Show your working clearly.

$$k = \dots\dots\dots$$

(4 marks)**Q45****PRACTICE**

Practice · Hard

CH 2 · TOPIC 2 3 marks

15 $\sqrt{2} \div \frac{8^3}{16^{\frac{3}{2}}} = 2^n$

Work out the value of n . Show your working clearly.

(3 marks)

PAST PAPERS

Q46**PAST PAPER**

Edexcel 4MA1 • January 2020 • Paper 2HR • Q21

CH 2 • TOPIC 2 7 marks

21 (a) Simplify fully $\frac{10x^2 + 23x + 12}{4x^2 - 9}$

$$2^{2y} \times 2^{3y+2} = \frac{8^{5y}}{4^n}$$

.....
(3)

(b) Find an expression for n in terms of y .

Show clear algebraic working and simplify your expression.

.....
(Total for Question 21 is 7 marks)

Q47**PAST PAPER**

Edexcel 4MA1 · Oct/Nov 2020 · Paper 1HR · Q21

CH 2 · TOPIC 2 5 marks

21 Given that $M = \frac{18^{4n} \times 2^{3(n^2-6n)} \times 3^{2(1-4n)}}{12^2}$

find the values of n for which $M = 2$

.....
(Total for Question 21 is 5 marks)

Q48**PAST PAPER**

Edexcel 4MA1 · January 2021 · Paper 2HR · Q19

CH 2 · TOPIC 2 7 marks**19 (a)** Simplify $8^2 \times \sqrt[3]{4^6}$ Give your answer in the form 2^a where a is an integer.

Show each stage of your working clearly.

.....
(3)Given that $n^{\left(-\frac{4}{5}\right)} = \left(\frac{1}{2}\right)^4$ where $n > 0$ (b) find the value of n . $n =$ **(Total for Question 19 is 7 marks)**

Q49**PAST PAPER**

Edexcel 4MA1 · May 2021 · Paper 1H · Q11

CH 2 · TOPIC 2 4 marks

11 $\sqrt{2} \times 16 = 2^x$

- (a) Find the value of x .
Show your working clearly.

$$x = \dots\dots\dots (2)$$

$$\frac{(11^{-6})^5}{11^4} = 11^n$$

- (b) Find the value of n .
Show your working clearly.

$$n = \dots\dots\dots$$

(Total for Question 11 is 4 marks)

Q50**PAST PAPER**

Edexcel 4MA1 · January 2022 · Paper 2HR · Q6

CH 2 · TOPIC 2 4 marks

- 6 (a) Write down the value of x^0

.....

Given that $2^{-3} \times 2^9 = 2^n$

- (b) find the value of n

$$n = \text{.....}$$

(1)

Given that $\frac{7^{206} \times 7^m}{7^{214}} = 7^{-3}$

- (c) find the value of m

$$m = \text{.....}$$

(Total for Question 6 is 4 marks)**Q51****PAST PAPER**

Edexcel 4MA1 · June 2022 · Paper 2HR · Q7

CH 2 · TOPIC 2 2 marks

7 $\frac{2^k}{4^n} = 2^x$

Find an expression for x in terms of k and n

$$x = \text{.....}$$

(Total for Question 7 is 2 marks)

Q52 **PAST PAPER** Edexcel 4MA1 · June 2022 · Paper 2H · Q24**CH 2 · TOPIC 2** **3 marks****24**

$$\frac{18 \times (\sqrt{27})^{4n+6}}{6 \times 9^{2n+8}} = 3^x$$

Express x in terms of n

Show your working clearly and simplify your expression.

 $x = \dots\dots\dots$ **(Total for Question 24 is 3 marks)**

Q53**PAST PAPER**

Edexcel 4MA1 · May 2023 · Paper 1HR · Q24

CH 2 · TOPIC 2**5 marks****24** Given that

$$2^n = 2^{x^2} \times 16^x \times 8$$

and

$$x > 0$$

find an expression for x in terms of n State any restrictions on n

(Total for Question 24 is 5 marks)**Q54****PAST PAPER**

Edexcel 4MA1 · June 2024 · Paper 2HR · Q7

CH 2 · TOPIC 2**2 marks**

$$7 \quad \frac{3^{-2} \times 3^5}{3^{10}} = 3^n$$

Find the value of n

$$n = \dots\dots\dots$$

(Total for Question 7 is 2 marks)

Q55**PAST PAPER**

Edexcel 4MA1 • June 2024 • Paper 2H • Q23

CH 2 • TOPIC 2 3 marks

23 Simplify $\frac{30 \times 25^{2x+7}}{\sqrt{180} \times (\sqrt{5})^{4x+9}}$

Give your answer in the form 5^w where w is an expression in terms of x
Show each stage of your working clearly.

(Total for Question 23 is 3 marks)

Q56**PAST PAPER**

Edexcel 4MA1 • November 2024 • Paper 2H • Q14

CH 2 • TOPIC 2 3 marks

14 Given that $\frac{3^{2n+3}}{3^4} = 3^3 \times 3^{1-2n}$

find the value of n
Show your working clearly.

$n =$

(Total for Question 14 is 3 marks)

Q57**PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 2HR • Q6

CH 2 • TOPIC 2**5 marks**

- 6 (a) Write down the value of
- 5^0

.....
(1)

$$\frac{5^9 \times 5^{-3}}{5^{-2}} = 5^k$$

- (b) Find the value of
- k

 $k =$

- (c) Simplify fully
- $(2d^4e^5)^3$

.....

(Total for Question 6 is 5 marks)**Q58****PAST PAPER**

Edexcel 4MA1 • November 2025 • Paper 1H • Q8

CH 2 • TOPIC 2**2 marks**

$$8 \quad \frac{8^{-2} \times 8^9}{8^{10}} = 8^n$$

Work out the value of n $n =$ **(Total for Question 8 is 2 marks)**

05 Where You Will Use It

Indices are the language of later topics. Learn to write everything as x^n .

WHEN YOU SEE → YOU DO

“in the form ax^n ”

$$\rightarrow \sqrt{x} = x^{\frac{1}{2}}, \frac{1}{x^2} = x^{-2}$$

differentiate $\frac{3}{x^2}$ or $\sqrt{x}$

→ rewrite as powers first

Differentiation

Rewrite first: $\frac{3}{x^2} = 3x^{-2}$,
 $\sqrt{x} = x^{\frac{1}{2}}$

Surds

$\sqrt{a} = a^{\frac{1}{2}}$ — the same laws apply.

Indices

Standard form

$$10^m \times 10^n = 10^{m+n}$$

Growth & decay

Compound interest: $P \times 1.05^n$

WORKED MODEL • WRITE IN THE FORM ax^n

$$\frac{5}{\sqrt[3]{x^2}}$$

$$1 \quad \sqrt[3]{x^2} = x^{\frac{2}{3}}$$

bottom is the root

$$2 \quad \frac{5}{x^{\frac{2}{3}}} = 5x^{-\frac{2}{3}}$$

flip = negative

ANSWER $5x^{-\frac{2}{3}}$

IDEA 05

Practise: Where You Will Use It

Questions written to practise exactly this idea.

Q59 DR ESLAM Differentiation: rewrite first

CH 2 • TOPIC 2 3 marks

Write each in the form ax^n : (a) $\frac{1}{x^3}$ (b) $\sqrt{x}$ (c) $\frac{5}{\sqrt[3]{x^2}}$

Q60 DR ESLAM Surds are powers

CH 2 • TOPIC 2 2 marks

Write $\sqrt{x} \times x^2$ as a single power of x .

Answer _____

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|----------------------------|---|
| 01 | Four laws | $\times$ add $\cdot \div$ subtract $\cdot ()^n$ multiply $\cdot$ everything inside gets the power. |
| 02 | Zero & negative | $a^0 = 1$. $a^{-n} = \frac{1}{a^n}$ — negative means flip. |
| 03 | Fractional | Bottom = root, top = power. Flip $\rightarrow$ Root $\rightarrow$ Power. |
| 04 | Solving | Unify the base (primes) $\rightarrow$ one power each side $\rightarrow$ equate powers. |
| 05 | Use it | Differentiation, surds, standard form, growth. |

BOTTOM LINE Same base $\rightarrow$ use the laws $\rightarrow$ negative = flip, fraction = root $\rightarrow$ to solve, unify the base and equate the powers.

Answers

Algebraic Roots & Indices • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • THE FOUR LAWS

- | | | |
|------------|--|--|
| Q1 | Dr Eslam • One law each | (a) x^8 (b) y^5 (c) x^{12} |
| Q2 | Dr Eslam • Numbers inside the bracket | $16x^{12}y^4$ |
| Q3 | Dr Eslam • Numbers, then letters | $\frac{3x^4}{y^3}$ |
| Q4 | Practice • Easy | (a) m^2 (b) $5x^6y^4$ |
| Q5 | Practice • Easy | a^4 |
| Q6 | Practice • Medium | $10a^5b^4$ |
| Q7 | Practice • Hard | $a = 2, b = 5$ |
| Q8 | Edexcel 4MA1 • January 2020 • Paper 2H • Q1 | (a) x^7 (b) 7^9 |
| Q9 | Edexcel 4MA1 • November 2021 • Paper 2H • Q1 | (a) $m = 9$ (b) $n = 21$ (c) $p = 4$ |
| Q10 | Edexcel 4MA1 • January 2022 • Paper 1H • Q1 | (a) a^{11} (b) w^{12} (c) $64x^{10}y^6$ (d) $t = \sqrt[3]{c + 8v}$ |
| Q11 | Edexcel 4MA1 • May 2023 • Paper 1HR • Q6 | (a) m^7 (b) $n = 8$ (c) $9x^{12}y^{16}$ |
| Q12 | Edexcel 4MA1 • November 2023 • Paper 2H • Q4 | (a) $x = 3.5$ (b) h^{12} (c) $16g^{12}k^{20}$ (d) $n = 14$ |
| Q13 | Edexcel 4MA1 • June 2025 • Paper 1H • Q4 | (a) $m = 12$ (b) $n = 5$ (c) $125a^{12}r^6$ |

IDEA 02 • ZERO AND NEGATIVE POWERS

- Q14** Dr Eslam • Zero and negative (a) 1 (b) $\frac{1}{8}$ (c) $\frac{9}{4}$
- Q15** Dr Eslam • Negative = flip x^{-4}
- Q16** Practice • Easy m^{-10}
- Q17** Practice • Easy 1
- Q18** Practice • Easy w
- Q19** Practice • Medium (a) $x = \pm 7$ (b) $\frac{1}{8}$ (c) $27x^6$ (d) $p = \frac{w+16}{4}$
- Q20** Edexcel 4MA1 • January 2021 • Paper 2HR • Q4 (a) $81k^8$ (b) $7m^4n^6$
- Q21** Edexcel 4MA1 • June 2022 • Paper 1H • Q9 (a) 8 (b) $p = 11$ (c) $8k^6m^{12}$
- Q22** Edexcel 4MA1 • June 2023 • Paper 2HR • Q7 (a) 1 (b) $m = -6$
- Q23** Edexcel 4MA1 • May 2024 • Paper 1H • Q10 (a) 1 (b) $n = 6$ (c) $125a^{12}c^6$

IDEA 03 • FRACTIONAL POWERS

- Q24** Dr Eslam • Bottom is the root (a) 4 (b) $\frac{1}{3}$ (c) $\frac{27}{8}$
- Q25** Dr Eslam • Flip → Root → Power $\frac{16y^2}{x^4}$
- Q26** Practice • Medium $3x^4y^{\frac{3}{2}}$
- Q27** Practice • Hard (a) 1 (b) $\frac{5y}{8x^3}$
- Q28** Practice • Hard $\frac{y^{15}}{8w^6}$
- Q29** Practice • Hard (a) $8a^9$ (b) $1000x^3$
- Q30** Edexcel 4MA1 • January 2020 • Paper 1H • Q17 (a) $36 + 24\sqrt{12} + 48 = 84 + 48\sqrt{3} = 12(7 + 4\sqrt{3})$
(b) $\frac{t^{10}}{9a^8}$
- Q31** Edexcel 4MA1 • January 2021 • Paper 1HR • Q12 (a) $4e^5f^3$ (b) $\frac{10x-5}{12}$ (c) $k = -2$
- Q32** Edexcel 4MA1 • January 2021 • Paper 2H • Q11 $4t^4w^2$
- Q33** Edexcel 4MA1 • May 2021 • Paper 2H • Q14 $\frac{4y^5}{3x^2}$
- Q34** Edexcel 4MA1 • June 2023 • Paper 2HR • Q11 $\frac{m^9k^{15}}{125}$
- Q35** Edexcel 4MA1 • June 2023 • Paper 2H • Q15 (a) $8a^9$ (b) $1000x^3$ (c) $y = 2.9$
- Q36** Edexcel 4MA1 • November 2023 • Paper 2H • Q12 (a) $x = 8.5$ (b) $25p^4y^{16}$
- Q37** Edexcel 4MA1 • May 2024 • Paper 1HR • Q17 (a) $n = 15$ (b) $\sqrt{147} + 14$
- Q38** Edexcel 4MA1 • June 2025 • Paper 2H • Q15 $125a^{12}c^6$
- Q39** Edexcel 4MA1 • November 2025 • Paper 2H • Q13 (a) $\frac{25x^4}{y^{10}}$ (b) $n = m - 3p$

IDEA 04 · SOLVING: UNIFY THE BASE

- Q40** Dr Eslam • Unify the base $x = 3$
- Q41** Dr Eslam • Both sides need changing (a) $x = 2$ (b) $x = \frac{5}{2}$
- Q42** Practice • Hard $x = \frac{29}{20}$
- Q43** Practice • Hard $m = -\frac{4}{3}$
- Q44** Practice • Hard $k = -2$
- Q45** Practice • Hard $n = -\frac{5}{2}$
- Q46** Edexcel 4MA1 • January 2020 • Paper 2HR • Q21 (a) $\frac{5x+4}{2x-3}$ (b) $n = 5y - 1$
- Q47** Edexcel 4MA1 • Oct/Nov 2020 • Paper 1HR • Q21 $n = 5$ or $n = -\frac{1}{3}$
- Q48** Edexcel 4MA1 • January 2021 • Paper 2HR • Q19 (a) 2^{10} (b) $n = 32$
- Q49** Edexcel 4MA1 • May 2021 • Paper 1H • Q11 (a) $x = 4.5$ (b) $n = -34$
- Q50** Edexcel 4MA1 • January 2022 • Paper 2HR • Q6 (a) 1 (b) $n = 6$ (c) $m = 5$
- Q51** Edexcel 4MA1 • June 2022 • Paper 2HR • Q7 $x = k - 2n$
- Q52** Edexcel 4MA1 • June 2022 • Paper 2H • Q24 $x = 2n - 6$
- Q53** Edexcel 4MA1 • May 2023 • Paper 1HR • Q24 $x = \sqrt{n+1} - 2, n > 3$
- Q54** Edexcel 4MA1 • June 2024 • Paper 2HR • Q7 $n = -7$
- Q55** Edexcel 4MA1 • June 2024 • Paper 2H • Q23 5^{2x+10}
- Q56** Edexcel 4MA1 • November 2024 • Paper 2H • Q14 $n = \frac{5}{4}$

IDEA 05 · WHERE YOU WILL USE IT

Q59 Dr Eslam · Differentiation:
rewrite first

$$(a) x^{-3} \quad (b) x^{\frac{1}{2}} \quad (c) 5x^{-\frac{2}{3}}$$

Q60 Dr Eslam · Surds are powers

$$x^{\frac{5}{2}}$$